

Average Examination Marks and Regression to the Mean

Gentlemen,

In a recent paper in Applied Statistics, Krzanowski, Mead and Thorne (1985) propose a mixed model for the analysis of marks obtained by students sitting a number of papers as part of the requirement for an honours degree. If all papers are assumed to be of equal standard this model may be simplified to the random effects model

$$Y_{ij} = \mu + \beta_j + \varepsilon_{ij} \quad (i = 1, \dots, m; j = 1, \dots, n)$$

where Y_{ij} is the examination mark of the j^{th} student in the i^{th} examination and $\beta_j \sim (0, \sigma_s^2)$ independently of $\varepsilon_{ij} \sim (0, \sigma^2)$.

A similar model has been considered by a number of authors in the field of clinical trials where patients may be graded for severity of some condition on the basis of a number of clinical measurements in a manner analogous to that of grading degree level students. Davis (1976) provides a useful review.

It is well known that patients in extreme groups selected on this basis will exhibit regression to the mean. For example if patients are chosen for a clinical trial because on the basis of a number of measurements their blood cholesterol is high, then they may be expected as a group to have lower blood cholesterol when subsequently measured.

Gardner and Heady (1973) consider the relationship between the 'true' mean level for such an extreme group and the measured mean level at the time of selection. If their results are transferred to the model above (which requires the assumption that β_j and ε_{ij} are normally distributed) and some fixed average mark criterion k is used for awarding first class degrees then we have

$$E[\bar{Y}_{.j} | \bar{Y}_{.j} > k] = \mu + \sqrt{(\sigma_s^2 + \sigma^2/m)} v\{(k - \mu)/\sqrt{(\sigma_s^2 + \sigma^2/m)}\}$$

and

$$E[\beta_j | \bar{Y}_{.j} > k] = \mu + \{\sigma_s^2/\sqrt{(\sigma_s^2 + \sigma^2/m)}\} v\{(k - \mu)/\sqrt{(\sigma_s^2 + \sigma^2/m)}\}$$

where $v(z) = \phi(z)/\{1 - \Phi(z)\}$ and $\phi(z)$ is the ordinate and $\Phi(z)$ the distribution function of the standardised Normal distribution. Clearly if σ^2/m is small the two expressions are similar and in general as m increases the two expressions approach the same limit

$$\mu + \sigma_s v\{k - \mu\}/\sigma_s\}$$

It may be shown that $E[\bar{Y}_{.j} | \bar{Y}_{.j} > k]$ is a decreasing function of m and $E[\beta_j | \bar{Y}_{.j} > k]$ is an increasing function of m . Thus, as the number of examinations increases the observed mean mark for first class students will decrease whereas the 'true' mean level of such students will increase.

An analogy with Bayesian statistics is instructive here. The distribution of β_j over all students may, for any given student, be regarded as a prior distribution which has progressively less and less relevance to his or her grade as the number of assessments increases until eventually it is appropriate to estimate β_j using $\bar{Y}_j$ alone. In the spirit of such an approach, an alternative to the rescaling of degree divisions, is one of rescaling marks using the estimated distribution of β_j . Expressed like this the issue facing an examination board may be put quite starkly. Is it appropriate in grading a student to take such external information into account? The extent to which Krzanowski's, Mead's and Thomas' scheme is adopted may reflect the answers of individual boards to this question.

Stephen Senn, Dundee College of Technology
(Circa 1985)

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